

Central Research Facility

Indian Institute of Technology Delhi

Ashwini K. Agrawal
Professor & Head

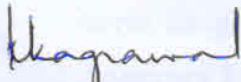
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Ref.: IITD/TT/AKA/CFC04
31 July 2018

Re: Minutes of CFC for Approval

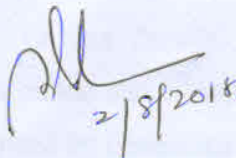
I am enclosing the minutes of the Central Facilities Committee meeting held on 25th June, 2018.

Kindly approve the minutes so that the listed activities may be started at the earliest.



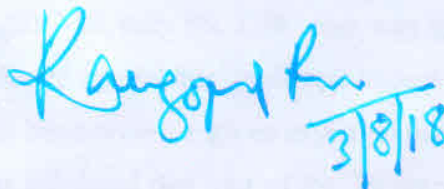
Ashwini K. Agrawal
(Head, CRF)

DD(S&P) Chairperson, CFC


2/8/2018

Please share these minutes with all faculty members.

Director


3/8/18

Head. ~~CRF~~, CRF

Indian Institute of Technology Delhi

CENTRAL RESEARCH FACILITY

Minutes of Meeting of Central Facility Committee (CFC) held on 25th June 2018 at 3:00 PM in the Board Room

Members present: Profs. M. Balakrishnan (DDS&P and Acting DD(O)), B. R. Mehta (Dean R&D), N. G. Ramesh, Amit Gupta, Shalini Gupta, Manidipa Banerjee, Pankaj Srivastava (Assoc. Head, CRF) and Ashwini Agrawal (Head, CRF)

The following matters were discussed and decisions taken for further action:

1. Head CRF presented the utilization of funds approved from MHRD special grant for equipment of CRF facilities. It was informed that 100% of sanction grant of Rs. 37.4 crore has been utilized and the orders have been placed to the vendors. The list of equipment along with the booked cost is provided in **Annexure 1**.
2. It was also informed that out of a loan of Rs. 65 crores approved from HEFA, the equipments worth Rs. 22.6 crore have been processed. The files are currently with the stores for placing the order. The details of equipment being processed are shown in **Annexure 2**.
3. It was also decided that from the remaining HEFA loan amount available for CRF facilities, some of the equipment remaining from the approved list of earlier surveys will be processed for purchase (**Annexure 3**), and subsequently, the equipment proposals received by various faculty groups will be considered. For the later (proposals received), an online survey will be conducted to evaluate their usability by faculty community followed by comments from an expert committee.
4. Head CRF also presented the utilization of grant received from IRD for operation of the facilities. It was informed that out of a total sanction budget of Rs. 2.63 crore for the year 2017-18, only Rs. 1.34 crore was spent. The remaining grant of Rs. 1.29 crore will be utilized against the approved budget for the current financial year. The expenditure details head-wise are given in **Annexure 4**.
5. It was informed that, out of Rs. 20 crores (Non-recurring grant) approved by IRD two years back, Rs. 2.0 crores was spent towards purchase of a new FESEM with accessories. It was decided that the remaining balance of Rs. 18 crores will be transferred back to IRD account. This saving is due to transfer of all equipment purchases to MHRD special grant as mentioned in point 1 above.

6. Head CRF informed that the system for charging user fee for all CRF facilities was successfully implemented from August 2017. The online booking software was suitably modified to keep the accounts of all users. The system is running perfectly. The revenue generated from internal as well as external users is shown in **Annexure 5**.
7. The committee suggested that the budgeted grants and earnings (from within and outside sources) should be separately listed and compared, in future reports, to allow assessment of the utilization of resources and the true revenues.
8. The committee also suggested that the software should be modified to project percentage utilization of various facilities and the major user departments.
9. Head CRF also informed that the voluntary feedback system has been implemented for the online booking, however, the committee decided that the feedback system should be made mandatory for users on next log-in into the system.
10. CFC agreed to accept the following proposals for adoption as per the CRF policy.
 - (i) HRTEM of Deptt. of Physics, Coordinator/Owner Prof. B. R. Mehta/AK Ganguly
 - (ii) FTIR of Deptt. of Chemical Engg., Coordinator/Owner Prof. Sreedevi Upadhyayula
 - (iii) Bio AFM from KSBS, Coordinator/Owner Prof. Bishwajit Kundu
 - (iv) GC-MS from DBEB, Coordinator/Owner Prof. Shaikh Z. Ahammad
 - (v) SQUID from Deptt. of Physics, Coordinator/Owner Prof. Rantanmala Chatterjee
 - (vi) FPLC from KSBS, Coordinator/Owner Prof. Tapan K. Chaudhuri
 - (vii) AFM from Deptt. of Applied Mechanics, Coordinator/Owner Prof. Sitikantha Roy
 - (viii) FACS from KSBS, Coordinator/Owner Prof. Archana Chugh
 - (ix) 3D Printer from Central Workshop, Coordinator/Owner Prof. Ravi Kumar
 - (x) Laser Cutter from Central Workshop, Coordinator/Owner Prof. Ravi Kumar

The above facilities will be made available for at least 3 days in a week for all CRF users through online booking and CRF will invest in maintenance and running of the facility as per CRF policy. Also, the coordinators of the facilities will be their respective owners or their representatives.

11. Head CRF also proposed that many facilities available with different deptt./faculty members have been included under CRF as shared facilities. These will be available for

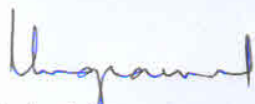
one day/week. As per the CRF policy, the owners of these facilities will be credited the testing fee on actual use basis. There will be no other liability of CRF towards these facilities. The list of these facilities is given in **Annexure 6**.

12. Head CRF informed the committee that a proposal is being discussed to bring a set of fabrication equipment from central workshop/other such facilities under CRF on adoption basis and to appoint couple of staff to assist students with their fabrication jobs. These equipment/jobs would be available to all users through online booking. The committee agreed to implement the proposal at the earliest for the benefit of the entire community. To start with, 3D printer and laser cutting machines have been offered by the Head Workshop for adoption by CRF.
13. Dean R&D informed the desire of the Director to rationalize testing grants being provided to research students under CRF and NRF. The committee strongly recommended that equipment available under NRF for use of all students be booked online through CRF online booking system. This will not only rationalize the testing grant being provided by the institute but also make the booking process more transparent and accountable. Dean R&D was asked to take this matter with NRF.
14. Dean R&D informed the committee that IRD has been providing matching grants to new faculty on the condition that the major equipment be shared under CRF. Also, the other specialized/minor equipment bought under these scheme, which need not be available through online booking, can be listed on the CRF website so that users may approach the concerned faculty for occasional use as per their need.
15. Dean R&D also informed the committee that Institute has sanctioned several grants under grand challenge scheme funded through special grant from MHRD. As per the terms of the scheme, the ownership of all the equipment to be bought under these projects will automatically be of CRF and these equipment should be made available to all users through online booking.
16. The committee recommended that users be advised to put acknowledgement for use of CRF facilities in their research papers/thesis. To promote this, special incentive scheme (in terms of discount or special testing grant) may be formulated to those who publish papers while acknowledging the CRF facilities.
17. The committee approved a budget of Rs. 5.10

18. crores under MI01116 for the year 2018-19 for the existing and newly adopted facilities as per the requirements sent by the various facility coordinators. The details of the budget head-wise are given in **Annexure 7**.
19. The committee approved the creation of additional project staff positions for CRF Delhi and Sonipat campuses for running and maintenance of facilities. These are given in **Annexure 8**.
20. The committee suggested that the requirement of accommodation for senior CRF staff posted at Sonipat may be assessed and communicated to FITT with instructions to reserve staff apartments available at Sonipat for CRF staff.
21. The committee discussed about the future requirement of permanent CRF staff. It was decided that the advertisements for permanent positions for CRF should be primarily based on qualifications (M.Tech. and PhD) and with requirement of relevant experience. The requirement of experience at a certain grade pay should be waived off for CRF staff.
22. The committee approved to provide an institute staff as Storekeeper cum Accountant for CRF from institute and decided that the inventory should be in the name of CRF Store Keeper to maintain accountability and continuity.
23. Head CRF informed the committee that it is usually difficult to deal with Institute staff posted for CRF work as they report to various departmental heads. It was recommended that such staff should either be transferred to CRF with their reporting to Head CRF or be removed from CRF facilities. This matter may be taken up by DDS&P with respective Heads and resolved.
24. High tension wire issue in CRF Sonipat was discussed. The committee approved that the help of suitable expert be taken on consultation basis and the CRF building be shielded accordingly for smooth installation and operation of the instruments.
25. The CFC decided that students from Haryana based Universities and colleges will be charged user fee equivalent to that charged to IIT Delhi students. This is as per the agreement signed by IIT Delhi with Haryana government.
26. CFC requested the DDS&P to provide some office space for CRF staff and for storage of files at the earliest. Also, CFC agreed to provide some additional space for installation of new equipment within IIT Delhi Hauz Khas campus. In addition to this, space for installation of new FESEM, which has already arrived, will be made available.

27. It was decided that there must be a proper power backup for each facility. In this regard, CFC agreed to provide a DG supply for all major CRF associated instruments as per their requirement/specification.
28. Some of the issues, as mentioned below, faced by CRF were discussed. The committee recommended that these issues are important for smooth functioning of CRF and Head CRF should seek help of DD (O) in resolution of issues.
- The committee was informed that ACs of newly built TEM/Microscopy Lab (behind CCD) have been malfunctioning for long and need replacement. The request in this regard was sent several times to relevant authorities, however no action was taken. It was decided to take-up the issue with DD (Operation).
 - There is delay in processing of bills both through the institute accounts and IRD.
 - The institute allows discarding of import papers by stores after five years. Absence of import papers like bill of lading etc. creates a serious problem in re-exporting of parts/machines for repairs to the manufacturing company. It is important to keep the files of major equipment for at least 10 years or else scanned before disposal. Also, the files may be sent to buyer/Head after 5 years for safe-keeping.

The meeting ended with vote of thanks to the Chair.



Ashwini K. Agrawal
Head CRF



(M. Balakrishanan)
Chairman, CFC & DD(S&P)



Director (For Approval)

All CFC Members

Appendix - I

Equipment Required for the Proposed NIPD Clinic & IED Project
(APR 2014)

S.No	Material Description	Quantity	Unit Price (Rs.)
1	WINDSHIELD	1	100000
2	WINDSHIELD	1	100000
3	WINDSHIELD	1	100000
4	WINDSHIELD	1	100000
5	WINDSHIELD	1	100000
6	WINDSHIELD	1	100000
7	WINDSHIELD	1	100000
8	WINDSHIELD	1	100000
	Total		800000

Annexure - 1

Equipments Purchased from Special MHRD Grant & IRD Project (RP03154G)

S. No.	Name of Instrument	Budget Code	Amount (Crores)
1	XPS	PLN03SF/CR	7.65 crores
2	SAXS	PLN03SF/CR	4.4 crores
3	XRD system	PLN03SF/CR	3.40 crores
4	HRTEM	PLN03SF/CR	13.4 crores
5	UTM	PLN03SF/CR	3.6 crores
6	Thermal Series	PLN03SF/CR	2.2 crores
7	Rheometer	PLN03SF/CR	2.05 crores
8	FTIR	PLN03SF/CR	0.7 crores
		Total	37.4 crores

Annexure-2

Equipments Purchase in Progress from HEFA Funds

S.No	Name of Instrument	Budget Code	Amount (Crores)
1	Ion Milling System	PLN01/HF03	3.2 crores
2	Femto Second Laser Cutting System	PLN01/HF03	5.0 crores
3	STM	PLN01/HF03	8.4 crores
4	PPMS	PLN01/HF03	6.0 crores
		Total	22.6 crores

Annexure - 3

Equipment Approved from Committee for Purchase under CRF

S. No.	Equipment
1.	Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS)
2.	Porosimeter
3.	Particle Size Analyser- wet and dry
4.	Confocal Optical Microscope
5.	High Resolution Stylus Profilometer
6.	Impedance Spectroscopy
7.	Ultra Centrifuge

Annexure - 4

Statement of Expenditure for "MI01116" (2017-18)

S. No.	Sanctioned Heads	Total Approved Budget in Lakhs (2017-18)	Expenditure in Lakhs (2017-2018)	Balance in Lakhs
1.	Non-Recurring Grant	77.67	37.58	40.09
2.	Recurring Grant	100	46.21	53.79
3.	Staff Salary	85.33	50.25	35.07
Total (Lakhs)		263	134	129

Annexure - 5

Revenue Generated from Internal User (Aug 2017 – Mar 2018)

S. No.	Instrument Name	Samples	Users No.	Amount (Rs.)	Start Date
1	SEM	3931	1156	4,28,955	Feb-2016
2	EDX	346	119	95,800	Feb-2016
3	Micro RAMAN	1107	283	78,200	Feb-2016
4	Solid State NMR	9	9	11,400	Mar-2016
5	HRTEM	511	178	1,15,320	Apr-2016
6	Cryo HRTEM	15	15	6,000	May-2016
7	XRD	103	98	1,750	May-2016
8	FESEM	580	211	90,775	May-2016
9	FESEM + EDX	277	114	83,550	May-2016
10	TEM CRF	432	264	61,725	Aug-2016
11	TEM DMSE	493	299	59,125	Aug-2016
12	LN2	14370	441	2,78,342	July-2017
13	Glass Blowing	433	282	2,65,600	Sept-2017
14	NMR 400 MHz	2854	1946	64,672	Sept-2017
15	NMR 300 MHz	1708	953	2,353	Oct-2017
16	ICPMS (Liq. Sample)	243	32	87,137	Oct-2017
17	ICPMS (Solid)	22	4	47,800	Oct-2017
18	Ultramicrotome	52	27	2,860	Nov-2017
19	FTIR	61	29	17,500	Dec-2017
20	MALDI - MS	53	27	24,750	Jan-2018
21	NMR 500 MHz	90	67	2,740	Feb-2018
22	TEM GRID	17	13	5,700	Feb-2018
Total		27707	6567	18,32,054	

Revenue Generated from Internal User (Apr 2018 – June 2018)

S. No.	Instrument Name	Samples	User No.	Amount (Rs.)	Start Date
1	SEM	1286	405	2,38,780	Feb-2016
2	EDX	164	57	53,750	Feb-2016
3	Micro RAMAN	119	40	26,400	Feb-2016
4	Solid State NMR	13	13	15,200	Mar-2016
5	HRTEM	89	37	48,860	Apr-2016
6	Cryo HRTEM	0	0	0	May-2016
7	XRD	0	0	0	May-2016
8	FESEM	23	11	12,650	May-2016
9	FESEM + EDX	0	0	0.00	May-2016
10	TEM CRF	122	74	34,650	Aug-2016
11	TEM DMSE	52	31	11,275	Aug-2016
12	LN2	4995	149	1,28,854	July-2017
13	Glass Blowing	101	64	99,675	Sept-2017
14	NMR 400 MHz	1683	1138	32,625	Sept-2017
15	NMR 300 MHz	9	8	431	Oct-2017
16	ICPMS (Liquid)	206	30	67,399	Oct-2017
17	ICPMS (Solid)	91	18	14,400	Oct-2017
18	Ultramicrotome	8	4	800	Nov-2017
19	FTIR	55	26	17,250	Dec-2017
20	MALDI - MS	71	29	23,875	Jan-2018
21	NMR 500 MHz	27	21	1,680	Feb-2018
22	TEM GRID	54	26	17,900	Feb-2018
23	Atom Probe	0	0	0	April -2018
24	SPRS	0	0	0	April -2018
25	HRTEM	52	30	27,500	May-2018
26	SQUID	21	19	20,400	May-2018
27	ESI MS	22	13	17,950	May-2018
28	FACS	14	4	4,200	June-2018
29	Bio AFM	0	0	0	June-2018
Total		9277	2247	9,16,504	

Revenue Generated from External Users (Apr 17- June 18)

S. No.	Instrument Name	Amount (Rs.)
1	FTIR	4,500
2	HRTEM	36,496
3	SEM	4,68,750
4	TEM	47,400
5	ICP-MS	74,050
6	Ultramicrotome	12,000
Total		6,43,196

Fund Transferred from Internal Faculty to CRF

S. No.	Faculty Name (Emp. Code)	Amount (Rs.)	Date
1	Dr. Anirban Das (11746)	2,000	19-09-2017
2	Dr. Sourabh Ghosh (16211)	30,000	27-10-2017
3	Dr. Leena Nebhani (16606)	50,000	27-10-2017
4	Dr. Preeti Srivastava (16340)	5,000	01-11-2017
5	Dr. Sourabh Ghosh (16211)	30,000	06-12-2017
6	Dr. A.K. Ghosh (15420)	10,000	07-11-2017
7	Dr. Naseer Hussain (16721)	5,000	14-11-2017
8	Dr. B. S. Butola (16180)	20,000	08-05-2018
9	Dr. Arun Kumar (16307)	20,000	28-05-2018
10	Dr. Ravi Shankar (15784)	5,000	09-06-2018
11	Dr. V. Ramgopal Rao (16653)	3,000	28-05-2018
12	Dr. V. Ramgopal Rao (16653)	3,000	23-02-2018
13	Dr. Viresh Dutta (15485)	25,000	15-02-2018
14	Dr. Naseer Hussain (16721)	7,000	18-12-2017
Total		2,15,000	

Annexure - 6

Facilities Proposed for Sharing in CRF

S. No.	Instrument	Department	Faculty in-charge
1	Antenna Facility	CARE	Prof. Ananjan Basu
2	Contact Angle measurement system	DMSE	Prof. Leena Nebhani
3	Atomic Absorption Spectrometer	DBEB	Prof. Ziauddin A. Shaikh
4	FPLC	DBEB	Prof. Saroj Mishra
5	Gas Chromatography (GC)	DBEB	Prof. Ashish Mishra
6	HPLC 1100 Series	DBEB	Prof. A.K. Srivastava
7	HPLC 1200 Series	DBEB	Prof. A.K. Srivastava
8	HPLC 1260 Series	DBEB	Prof. A.K. Srivastava
9	Elemental Analyser	DBEB	Prof. Ziauddin A. Shaikh
10	Fluorimeter	DBEB	Prof. Ziauddin A. Shaikh
11	XRD	Textile	Prof. Bhanu Nandan
12	SPR Spectroscope	Chemical	Prof. Shalini Gupta

Annexure - 7

Proposed Budget of CRF for 2018-19

S. No.	Facility	Non-Recurring	Spare Part	Recurring	Staff
1	SEM	8	0	18	11
2	ESI-LC-MS	0	2	8	5.27
3	Raman	0	20	4.5	5.27
4	FESEM + EDX	10	0	6.5	5.27
5	Solid State NMR	0	6.5	13	5.27
6	TEM-CRF	0	1.5	9	7
7	TEM-DMSE	1	0	7	5.63
8	NMR	3	0	9.5	5.27
9	Glass Blowing	5	0	22.5	4
10	Liquid N2	0	0	3	8
11	ICP-MS	0	38	7	5.27
12	Cryo HR-TEM	0	4	8.5	5.63
13	Atom Probe	0	25	1	0
14	MALDI-MS	2.5	0	8	5.27
15	Ultramicrotomy	0	0	2	0
16	FACS	0	0	10.5	0
17	SQUID	18	0	5	5.27
18	HR-TEM	0	50	4.5	7
19	BIO-AFM	0	2	4	2.65
20	FTIR	1	4	1.5	1.2
21	GC-MS	0	2.5	7	5.27
22	AFM	0	0	14	2.65
23	3D Printer	0	0	1	0
24	Laser Cutter	0	0	1	5.27
25	Senior Staff	0	0	0	22.5
	Total	48.5	155.5	176	129.96
	Grand Total	509.96			

Annexure - 8

Staff Approval Required for Sonipat Campus

S. No.	Post	No.	Qualifications	Job-Profile
1	Pr. Project Scientist	5	Ph.D. with relevant experience	Administration, Maintenance and Management; Handling of TEM, Ion Milling System, XPS, PPMS and STM
2	Sr. Project Scientist	4	M.Tech. with hands-on experience	Handling of Femto Second Laser Cutting, UTM, FTIR, XRD, SAXS, Thermal Series and Rheometer
3	Jr. Project Assistant	1	ITI with experience	Technical Assistance
4	Lab Attendant	1	12th Pass	Office Paper Related Work

Staff Approval Required for Hauz-Khas Campus

S. No.	Post	No.	Qualifications	Job-Profile
1	Pr. Project Scientist	1	Ph.D. with relevant experience	Administration, Maintenance and Management
2	Sr. Project Scientist	8	M.Tech. with hands-on experience	Handling of newly adopted equipment SQUID, ESI-LC-MS, FESEM + EDX, Cryo HR-TEM, HR-TEM, Bio-AFM, AFM, GC-MS, 3D Printer and Laser Cutter